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GitHub Link	https://github.com/kiduuuu/MLA0405-Deep-learning/tree/main

SIMATS ENGINEERING

CO5 - ASSESSMENT TOOL 3

Technical Presentation

COURSE NAME:Deep Learning
SLOT : D

COURSE CODE:MLA0405
Faculty : Dr.Arul Raj A.M

COURSE OUTCOME COVERED

CO5: Students will be able to analyze and explain advanced generative deep learning models including Autoencoders, Deep Belief Networks, Boltzmann Machines, Deep Boltzmann Machines and Generative Adversarial Networks. They will be able to compare their architectures, explain their learning mechanisms, and apply suitable generative models to real-world problems. (BTL-3)

Course Outcome Weightage: 35%
Assessment Tool Weightage: 50%

Duration: 90 minutes
Mark : 25

Code of Conduct:

I (D.Chinna Moulali / 192425278) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *D.Chinna Moulali*

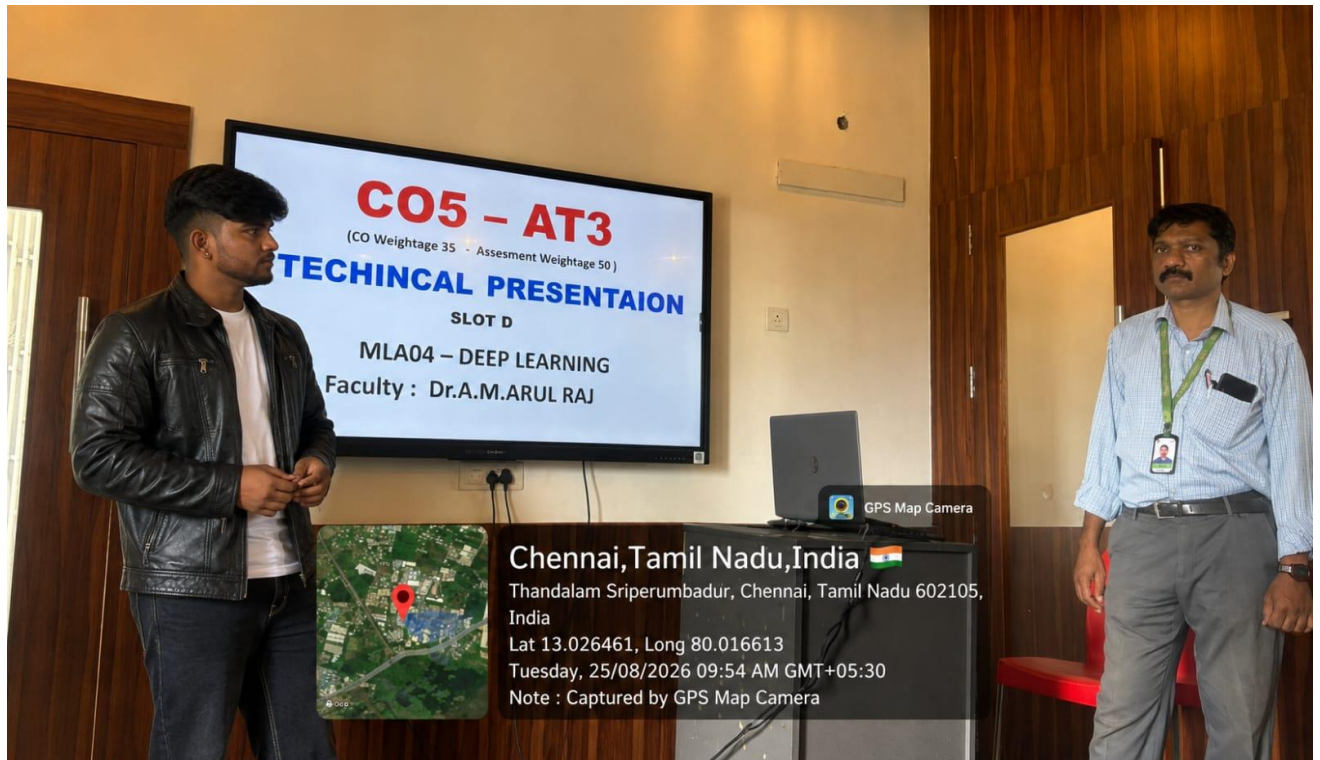
Name of the student :D.Chinna Moulali

Criteria	Understanding of Problem Context (2)	Application of Concepts (3)	Problem-Solving Approach (3)	Accuracy of Solution (2)	Clarity (1)	Total (10)
Weightage	20 %	30%	30%	20 %	10 %	100%
Q1						
Total						

Signature of the Faculty:

Total Marks:

PHOTO :



QUESTIONS: Stochastic Encoders, Decoders and Contractive Encoders

Prepare a technical presentation explaining Stochastic Encoders and Decoders and Contractive Encoders. Describe their working principles, objectives, robustness characteristics, and differences from standard autoencoders. Demonstrate a suitable real-world application for each approach.

PRESENTATION ANSWER:

Stochastic Encoders, Decoders and Contractive Encoders

Slide 1 – Title

Stochastic Encoders, Decoders and Contractive Encoders

- Advanced Autoencoder Architectures
- Focus: Working Principle, Objectives, Robustness and Applications

Slide 2 – Introduction to Autoencoders

- An **Autoencoder (AE)** is an unsupervised neural network used to learn compact representations of data.
- It consists of two main components:
 - **Encoder:** Converts input data into a latent representation.
 - **Decoder:** Reconstructs the original data from the latent representation.
- Objective: Minimize the difference between input and reconstructed output.

- Standard autoencoders generally learn a **deterministic mapping**.

Basic flow:

Input $\rightarrow$ Encoder $\rightarrow$ Latent Representation $\rightarrow$ Decoder $\rightarrow$ Reconstruction

Slide 3 – Stochastic Encoder

- A **Stochastic Encoder** does not produce a single fixed latent vector.
- Instead, it generates a **probability distribution** over the latent representation.
- Usually represented using:
 - Mean: μ
 - Variance: σ^2
- A latent sample is obtained from this distribution.
- This introduces controlled randomness into the encoding process.

Flow:

Input $x \rightarrow$ Encoder $\rightarrow \mu, \sigma \rightarrow$ Sampling $\rightarrow$ Latent z

Slide 4 – Working of Stochastic Encoder

1. Input data is provided to the encoder.
2. The neural network extracts important features.
3. The encoder produces parameters **μ and σ** .
4. A probability distribution is constructed.
5. A latent vector **z** is sampled from the distribution.
6. The sampled vector is passed to the decoder.

A common sampling method is the **reparameterization trick**:

$$z = \mu + \sigma \odot \epsilon$$

where ϵ is random noise sampled from a standard normal distribution.

Slide 5 – Stochastic Decoder

- A **Stochastic Decoder** models uncertainty during reconstruction.
- Instead of always producing one fixed output, it can represent a **distribution of possible outputs**.
- It is useful when multiple outputs are reasonable for the same input.
- The decoder learns the probability distribution of the reconstructed data.

Flow:

Latent $z \rightarrow$ Stochastic Decoder $\rightarrow$ Output Distribution $\rightarrow$ Reconstructed Sample

Slide 6 – Objectives of Stochastic Encoders and Decoders

- Learn meaningful latent representations.
- Model uncertainty in data.
- Generate new and realistic samples.
- Improve generalization.
- Handle noisy or incomplete information.
- Represent complex data distributions.
- Provide more flexible latent spaces than deterministic autoencoders.

Slide 7 – Robustness Characteristics

Stochastic autoencoders can provide robustness through controlled randomness.

- Less dependent on an exact latent representation.
- Can handle variations and noise in input data.
- Encourage smoother latent representations.
- Improve generalization to unseen samples.
- Useful for uncertain and probabilistic environments.

Example:

A slightly noisy image can still produce a meaningful latent representation and reconstructed image.

Slide 8 – Contractive Encoder

- A **Contractive Encoder** is designed to make the learned representation insensitive to small changes in the input.
- It adds a **contractive regularization penalty** to the training objective.
- The encoder learns representations that remain stable when the input is slightly perturbed.
- It is particularly useful for learning robust features.

Basic idea:

Small Input Change → Very Small Latent Change

Slide 9 – Working of Contractive Encoder

1. Input is passed through the encoder.
2. The encoder generates a latent representation.
3. The reconstruction error is calculated.
4. The sensitivity of the encoder to input changes is measured.
5. A contractive penalty is added to the loss.
6. The network learns a stable representation.

The objective can be represented as:

$$\text{Total Loss} = \text{Reconstruction Loss} + \lambda \times \text{Contractive Penalty}$$

The contractive penalty is commonly based on the **Frobenius norm of the encoder Jacobian**:

$$L = L_{\text{reconstruction}} + \lambda \|Jf(x)\|_F^2$$

where $Jf(x)$ represents how sensitive the encoder representation is to input changes.

Slide 10 – Robustness of Contractive Encoders

- Reduces sensitivity to small input perturbations.
- Learns stable and invariant features.
- Helps suppress irrelevant variations.
- Can improve performance with noisy data.
- Produces smoother representations around training examples.
- Useful for pattern recognition and feature extraction.

Slide 11 – Standard vs Stochastic vs Contractive Autoencoder

Feature	Standard AE	Stochastic AE	Contractive AE
Latent representation	Deterministic	Probabilistic	Deterministic
Randomness	Low/None	Introduced intentionally	Low/None
Main objective	Reconstruction	Reconstruction + distribution learning	Reconstruction + stability
Handles uncertainty	Limited	Excellent	Limited
Noise robustness	Moderate	Good	Good
Main strength	Compression	Generative representation	Robust features
Typical use	Dimensionality reduction	Data generation	Feature learning

Slide 12 – Real-World Application: Stochastic Autoencoder

Application: Medical Image Generation and Reconstruction

- Medical images can contain uncertainty and variations.
- A stochastic encoder learns a probability distribution for image features.
- The decoder can generate different plausible reconstructions.

- Useful for:
 - MRI image reconstruction
 - CT image enhancement
 - Missing-data reconstruction
 - Synthetic medical image generation

Example:

A partially corrupted MRI scan can be represented probabilistically and reconstructed into plausible versions.

Slide 13 – Real-World Application: Contractive Autoencoder

Application: Handwritten Character Recognition

- Handwritten characters can vary in size, angle and stroke thickness.
- A contractive encoder learns features that are less sensitive to these small variations.
- The resulting representation focuses on important character structures.
- This can improve classification and recognition.

Example:

Different handwritten versions of the digit “5” can produce similar latent representations even when their strokes differ slightly.

Slide 14 – Advantages

Stochastic Encoders/Decoders

- Model uncertainty effectively.
- Support data generation.

Contractive Encoders

- Learn robust representations.
- Reduce sensitivity to small perturbations.

Slide 15 – Limitations

Stochastic Approach

- Training can be computationally more complex.
- Random sampling can make optimization challenging.
- Requires careful probability-distribution design.

Contractive Approach

- Additional regularization increases training complexity.
- Selecting the regularization parameter λ is important.
- Excessive contraction may remove useful information.

Slide 16 – Conclusion

- **Standard Autoencoders** mainly focus on reconstructing input data.
- **Stochastic Encoders and Decoders** introduce probability and uncertainty into representation and reconstruction.
- **Contractive Encoders** focus on learning representations that are stable against small input variations.
- Stochastic models are especially suitable for **generation, uncertainty modeling and reconstruction**.
- Contractive models are suitable for **robust feature extraction and recognition**.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 Exemplary	Level 3 Proficient	Level 2 Developing	Level 1 Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand